ECE 310 Fall 2026
Lecture 14: Discrete-time Fourier Transform Properties

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Learning Objectives

After this lecture, you should be able to:

- Explain the relationship between the z-transform and the DTFT.
- Apply key properties of the DTFT to compute transform pairs and analyze signals in both the time-domain and frequency-domain.



Last Time: Discrete-time Fourier Transform

Discrete-time Fourier Transform

The discrete-time Fourier transform (DTFT) of a discrete-time signal $x[n]$ is denoted as $X_d(\omega)$: $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)$.

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X_d(\omega) e^{j\omega n} d\omega$$

The DTFT of the signal $x[n]$ exists if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$



Example: Computing the DTFT

Compute the DTFT of the following sequences.

$x[n] = \{1, \underset{\uparrow}{0}, 0, -1\}$

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= e^{j\omega} - e^{-j2\omega} = e^{-j\frac{1}{2}\omega}\left(e^{j\frac{3}{2}\omega} - e^{-j\frac{3}{2}\omega}\right)$$

$$= e^{-j\frac{1}{2}\omega}\left(2j \sin\left(\tfrac{3}{2}\omega\right)\right)$$

$x[n] = \{\underset{\uparrow}{1}, 2, 3\}$

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= 1 + 2e^{-j\omega} + 3e^{-j2\omega}$$

$x[n] = a^n u[n], \; a \in \mathbb{C}, \; |a| < 1$

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \left(ae^{-j\omega}\right)^n$$

$\sum_{n=0}^{\infty} b^n = \frac{1}{1-b}$ if $|b|<1$

$$= \frac{1}{1 - ae^{-j\omega}}$$

How are the z-transform and DTFT related?

Thus far, we have seen:

- The z-transform shows how much each geometric series z^n, $z \in \mathbb{C}$ contributes to a discrete-time signal.
- The DTFT shows how much each periodic complex exponential $e^{j\omega_0 n}$, $\omega_0 \in \mathbb{R}$, contributes to a discrete-time signal.

If $z = e^{j\omega_0}$, we see how much $e^{j\omega_0 n}$ is in $x[n]$ from $X(z)$ and $X_d(\omega)$

Let $z = e^{j\omega}$: $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \Big|_{z=e^{j\omega}}$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = X_d(\omega)$$

The DTFT is the z-transform evaluated on the unit circle!

Concept Check: z-transform and DTFT

Consider the following z-transforms:

Let $z = e^{j\omega}$

$$X_1(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}, \ |z| > \frac{1}{2}$$

$$X_2(z) = \frac{1 + z^{-1}}{1 - 2z^{-1} - 3z^{-2}}, \ |z| > 2$$

Determine the DTFT of each signal $X_1(\omega)$ and $X_2(\omega)$, if they each exist.

1. a) Does not exist b) $X_1(\omega) = \dfrac{1 - e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$ c) $X_1(\omega) = \dfrac{1 - j\omega}{1 - \frac{1}{2}j\omega}$

2. a) Does not exist b) $X_2(\omega) = \dfrac{1 + e^{-j\omega}}{1 - 2e^{-j\omega} - 3e^{-j2\omega}}$ c) $X_2(\omega) = \dfrac{1 + j\omega}{1 - 2j\omega - 3(j\omega)^2}$

DTFT of Periodic Signals

As we saw in the last lecture, the DTFT (and respectively the CTFT) can also be seen as an extension of the Fourier series. However, every periodic signal is not absolutely summable!



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But we would still like to work with periodic signals in the frequency domain, so what is a reasonable representation for the DTFT of a periodic signal? For example, the DTFT of $x[n] = e^{j\omega_0 n}, \omega_0 \in [0, 2\pi)$.



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$$X_d(\omega) = \begin{cases} \text{Non-zero}, & \omega = \omega_0 + 2\pi k, \ k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$



DTFT of Periodic Signals

Let $x[n] = e^{j\omega_0 n}$:

$$X_d(\omega) = \begin{cases} \text{Non-zero}, & \omega = \omega_0 + 2\pi k,\ k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$

Dirac delta:

$$\delta(t-a) = \begin{cases} \infty, & t = a \\ 0, & \text{else} \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t-a)\,dt = 1$$

$$\int_{-\infty}^{\infty} \delta(\omega-a) f(\omega)\,d\omega = f(a)$$

$$X_d(\omega) = c\,\delta(\omega-\omega_0)$$

$$x[n] = \frac{1}{2\pi}\int_0^{2\pi} X_d(\omega)\, e^{j\omega n}\,d\omega$$

$$e^{j\omega_0 n} = \frac{1}{2\pi}\int_0^{2\pi} c\,\delta(\omega-\omega_0)\, e^{j\omega n}\,d\omega$$

$$e^{j\omega_0 n} = \frac{c}{2\pi}\left(e^{j\omega_0 n}\right)$$

$$\rightarrow c = 2\pi$$

$$e^{j\omega_0 n} \xleftrightarrow{\mathcal{F}} 2\pi\,\delta(\omega-\omega_0)$$

Common DTFT Pairs

Signal $x[n]$	DTFT $X_d(\omega)$
$\delta[n]$	1
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \pi\delta(\omega)$
$a^n u[n]$	$\frac{1}{1-ae^{-j\omega}},\ 0 < \lvert a\rvert < 1$
$e^{j\omega_0 n}$	$2\pi\delta(\omega-\omega_0)$
$\cos(\omega_0 n)$	$\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$
$\sin(\omega_0 n)$	$-j\pi\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$
$\text{rect}\left(\frac{n-k}{L}\right)$	$\frac{\sin\left(\frac{L\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}e^{-j\omega k}$
$\text{sinc}(Ln)$	$\frac{\pi}{L}\text{rect}\left(\frac{\omega}{2L}\right)$
$\text{sinc}^2(Ln)$	$\frac{\pi}{L}\Delta\left(\frac{\omega}{2L}\right)$

DTFT Properties: 2π Periodicity

The DTFT of any discrete-time signal $x[n]$, where its DTFT exists, is 2π-periodic.

$$X_d(\omega) = X_d(\omega + 2\pi k),\ k \in \mathbb{Z}$$

$$X_d(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j(\omega + 2\pi k)n}$$

$$= \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}\, e^{-j2\pi kn}, \quad kn = m \in \mathbb{Z} \text{ (integer)}$$

$$= \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$= X_d(\omega)$$

DTFT Properties: Hermitian Symmetry

If $x[n]$ is real-valued with DTFT $X_d(\omega)$, then $X_d(\omega)$ is *Hermitian symmetric*. This means

$$X_d^*(\omega) = X_d(-\omega)$$

which is equivalent to

$$|X_d(\omega)| = |X_d(-\omega)| \text{ and } \angle X_d(\omega) = -\angle X_d(-\omega).$$

*Note: this is an "if and only if" condition!

This is an "if and only if" statement

$x[n]$ real $\longleftrightarrow$ $X_d(\omega)$ is Hermitian symmetric

even symmetry

odd symmetry

Even symmetry: ω^2, $|\omega|$, $\cos(\omega)$

Odd symmetry: ω, ω^3, $\sin(\omega)$

Example: Checking Hermitian Symmetry

Suppose the DTFT of a discrete-time signal $x[n]$ is given by

$$X_d(\omega) = \omega^2 e^{j\cos(\omega)}.$$

Determine whether $x[n]$ is real-valued.

Hermitian symmetry: $X_d^*(\omega) = X_d(-\omega)$ or $X_d(\omega) = X_d^*(-\omega)$

$X_d^*(\omega) = \omega^2 e^{-j\cos(\omega)}$

$X_d(-\omega) = (-\omega)^2 e^{j\cos(-\omega)}$

$= \omega^2 e^{j\cos(\omega)}$

$X_d^*(\omega) \neq X_d(\omega)$

→ $x[n]$ is complex-valued

Concept Check: Hermitian Symmetry and 2π-periodicity

The DTFT of a real-valued signal $x[n]$ is given by $X_d(\omega)$. Which of the following relationships are true? (Select all that are true)

(a) $X_d(0) = X_d(2\pi)$

(b) $X_d\left(\frac{\pi}{3}\right) = X_d\left(-\frac{\pi}{3}\right)$

(c) $X_d\left(\frac{\pi}{3}\right) = X_d^*\left(-\frac{\pi}{3}\right)$

(d) $X_d(\pi) = X_d(7\pi)$

(e) $X_d(\frac{2\pi}{3}) = X_d^*(\frac{4\pi}{3})$

Hermitian symmetry: $X_d^*(\omega) = X_d(-\omega)$

subtract 2π

$\equiv X_d\left(-\frac{4\pi}{3}\right)$

Other DTFT Properties

Convolution:

$$x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)H_d(\omega)$$

Time-shifting:

$$x[n-k] \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega k}X_d(\omega), \ k \in \mathbb{Z}$$

Frequency-shifting and Modulation:

$$e^{j\omega_0 n}x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega - \omega_0)$$

$$x[n]\cos(\omega_0 n) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}X_d(\omega - \omega_0) + \frac{1}{2}X_d(\omega + \omega_0)$$

Multiplication in time (windowing):

$$x[n]w[n] \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi}X_d(\omega) * W_d(\omega)$$



DTFT Properties Table

Property	Signal	DTFT
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(\omega) + bX_2(\omega)$
Time shifting	$x[n-k]$	$X_d(\omega)e^{-jk\omega}$
Frequency shifting	$e^{j\omega_0 n}x[n]$	$X_d(\omega - \omega_0)$
Modulation	$x[n]\cos(\omega_0 n)$	$\frac{1}{2}X_d(\omega - \omega_0) + \frac{1}{2}X_d(\omega + \omega_0)$
Time reversal	$x[-n]$	$X_d(-\omega)$
Conjugation	$x^*[n]$	$X_d^*(-\omega)$
Differentiation	$nx[n]$	$-j\frac{dX_d(\omega)}{d\omega}$
Convolution	$x[n] * h[n]$	$X_d(\omega)H_d(\omega)$
Windowing	$x[n]w[n]$	$\frac{1}{2\pi}X_d(\omega) * W_d(\omega)$
Hermitian symmetry	$x[n]$ real	$X_d^*(\omega) = X_d(-\omega)$
Parseval's relation	$\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$	$\frac{1}{2\pi}\int_{2\pi} \lvert X_d(\omega)\rvert^2 d\omega$

